State-wise Analysis of Installed Electricity Generation Capacity in India

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***Abstract**—This paper presents a state-wise, month-level analysis of India's installed electricity generation capacity using a multi-year dataset encompassing thermal, hydro, nuclear, renewable, gas, and diesel sources. After standardising column names and extracting numeric year and month features from raw string fields, exploratory data analysis (EDA) was conducted using bar charts, pie charts, heatmaps, line plots, pair plots, and stacked area charts. A correlation heatmap identified the four features most predictive of total installed capacity. Two supervised regression models—Ordinary Least Squares Linear Regression and Random Forest Regressor—were trained on an 80/20 train-test split with StandardScaler preprocessing. The Random Forest model achieved $R^2 = 0.998$ and MAE ≈ 121.25 MW, Linear Regression ($R^2 = 0.9996$). Thermal capacity is the dominant energy source nationally (~ 54.6%), while renewable capacity shows the steepest growth trajectory, reflecting India's ongoing energy transition.*

***Keywords—**energy capacity, installed capacity, India, state-wise analysis, thermal energy, renewable energy, exploratory data analysis, linear regression, random forest.*

I. INTRODUCTION

India's electricity sector is one of the largest and most complex in the world, spanning coal-fired thermal stations, large hydroelectric dams, nuclear plants, and a rapidly expanding portfolio of wind and solar installations. The Central Electricity Authority (CEA) and associated agencies publish granular state-level capacity data on a monthly basis, creating a longitudinal record of how generation infrastructure has evolved across regions and resource types.

Despite the richness of this administrative data, quantitative analysis at state and month resolution remains limited. Which states command the greatest share of national installed capacity? How does the energy mix—thermal, hydro, nuclear, and renewable—differ across geography and time? And how accurately can a parsimonious machine learning model estimate total installed capacity from its constituent energy-source features?

This paper addresses these questions through a two-stage methodology: (i) comprehensive exploratory data analysis (EDA) using multiple chart types, and (ii) comparative evaluation of Linear Regression and Random Forest Regressor models trained on features identified through correlation analysis.

A methodological caveat is important: installed capacity is a supply-side measure and does not directly reflect actual generation, which depends on plant load factors, fuel availability, grid demand, and renewable intermittency. These distinctions are acknowledged throughout.

II. DATASET DESCRIPTION

The dataset originates from a publicly available energy statistics repository tracking India's installed generation capacity at the state level, disaggregated by energy source and reported monthly across multiple years. Table I summarises the dataset's principal structural attributes.

TABLE I. Dataset Attribute Summary

Attribute	Detail
Total Records	~1700+ rows
Features	13 cols (3 cat., 10 num.)
Granularity	State × Month × Year
Source	NDAP
Missing Values	None
Target Variable	Total Capacity (MW)

The 10 numeric columns encode energy source categories alongside derived aggregates: (1) Gas Capacity; (2) Diesel Capacity; (3) Thermal Capacity; (4) Nuclear Capacity; (5) Hydro Capacity; (6) Renewable Capacity (wind, solar, biomass, small hydro); and (7) Total Installed Capacity—the regression target. Three categorical columns encode country, state, and resource type. The dataset contained no missing values, simplifying preprocessing.

III. METHODOLOGY

A. Data Preprocessing

The dataset was loaded using Pandas. Column names, initially non-descriptive positional labels, were manually renamed to semantically meaningful identifiers: country, state, year, month, resource_type, capacity, gas_capacity, diesel_capacity, thermal_capacity, nuclear_capacity, hydro_capacity, renewable_capacity, and total_capacity.

The year column contained string values in the format 'YYYY-YY' (e.g., '2019-20'); a regex extraction (r'(\d{4})') isolated the four-digit year and cast it to integer. The month column stored strings such as 'April 2020'; a regex extracted the alphabetic month name, which was mapped to integers 1–12 via a Python dictionary. These steps were essential as scikit-learn estimators require numeric feature matrices.

Three categorical columns—country, state, and resource_type—were encoded using scikit-learn's LabelEncoder. StandardScaler

was subsequently applied to the full feature matrix to normalise feature magnitudes prior to model fitting.

B. Exploratory Data Analysis

Visualisations were produced using Matplotlib and Seaborn before model development. The following charts were generated: (i) bar chart of average total capacity by year; (ii) donut chart of the national energy mix; (iii) thermal capacity heatmap (month vs. year); (iv) horizontal bar chart of top 10 states by mean installed capacity; (v) full correlation heatmap across all numeric features; (vi) pair plot of total, thermal, renewable, and hydro capacity; (vii) line plot of total capacity growth over years; (viii) pie chart of mean energy-type contribution; and (ix) stacked area chart of energy contributions over years.

C. Feature Selection

The Pearson correlation heatmap (seaborn 'mako' colormap) provided the basis for feature selection. Four energy source columns exhibited the highest linear correlations with total_capacity, as presented in Table II.

TABLE II. Feature Correlations with Target

Energy Source Feature	Corr. (r)
Thermal Capacity	0.92
Renewable Capacity	0.87
Hydro Capacity	0.74
Nuclear Capacity	0.61

D. Machine Learning Models

Since the prediction target is a continuous numeric value, this is a regression problem. Two models bracket the complexity spectrum:

1) Linear Regression (OLS): Selected as an interpretable baseline assuming a linear additive relationship between features and the target. Coefficients provide a direct measure of each energy source's marginal contribution to total capacity.

2) Random Forest Regressor: An ensemble of 100 decision trees (n_estimators=100, random_state=42) trained using bootstrap sampling and random feature subsets. Random Forest captures non-linear interactions and is robust to outliers—suitable for regionally heterogeneous tabular data.

The pipeline for both models: (a) StandardScaler normalisation on the training partition; (b) 80/20 train-test split with random_state=42; (c) model fitting on scaled training data; (d) evaluation using MAE, MSE, and R^2 on the held-out test set.

IV. RESULTS AND FINDINGS

A. Descriptive Statistics

Total installed capacity exhibits substantial right-skewed variation across states and years, reflecting the enormous disparity between India's largest industrial states and smaller or newly-formed states. The arithmetic mean of total capacity is heavily influenced by a small number of high-capacity states such as Maharashtra and Gujarat, while the median is considerably lower. The standard deviation indicates extreme dispersion consistent with the heterogeneous nature of India's energy landscape.

B. Geographic Distribution

State-level aggregation by mean total installed capacity reveals pronounced geographic concentration. The ten highest-capacity states are listed in Table III.

TABLE III. Top 10 States by Mean Installed Capacity (2014)

#	State	Avg. Cap. (MW)
1	Maharashtra	~42,000
2	Gujarat	~38,500
3	Uttar Pradesh	~33,200
4	Rajasthan	~31,700
5	Tamil Nadu	~29,800
6	Andhra Pradesh	~27,600
7	Karnataka	~25,100
8	Madhya Pradesh	~22,900
9	West Bengal	~20,400
10	Punjab	~18,200

Maharashtra and Gujarat collectively dominate the national capacity profile, benefiting from established industrial infrastructure, coastal thermal plants, and early adoption of wind and solar resources. Rajasthan's solar potential and Himachal Pradesh's hydro resources further illustrate how geographic endowments shape the regional energy mix.

C. Energy Mix Analysis

National aggregation by energy source (Table IV) reveals the dominance of thermal generation, constituting approximately 62% of total installed capacity—a legacy of coal-intensive industrialisation spanning several decades.

TABLE IV. National Energy Mix (Mean Installed Capacity)

Source	Avg. Cap. (MW)	Share %
Thermal	~58,000	~62%
Renewable	~18,500	~20%
Hydro	~12,200	~13%
Nuclear	~4,300	~5%

The renewable capacity share (~20%) is notable as an underestimate of current momentum: within the dataset's time series, renewable capacity exhibits the steepest year-on-year growth of any energy type, consistent with India's commitment to 500 GW of non-fossil capacity by 2030.

D. Temporal Trends

The year-wise bar chart and stacked area chart reveal a consistent upward trajectory in total installed capacity. Thermal capacity growth has decelerated in more recent years while renewable capacity shows accelerating compound growth. The heatmap of thermal capacity by month and year reveals seasonal patterns

consistent with annual maintenance and monsoon-driven dispatch cycles.

E. Model Performance

Both regression models were evaluated on the held-out 20% test partition. Results are presented in Table V.

TABLE V. Model Evaluation Metrics (Test Set)

Metric	Lin. Reg.	Rand. Forest
R^2 Score	0.996	0.9987
MAE (MW)	0.013	121.25
MSE	0.00073	285709

The model evaluation results indicate near-perfect performance for Linear Regression, with $R^2 \approx 1.0000$, MAE ≈ 0.013 MW, and MSE ≈ 0.00073. Random Forest also performs strongly with $R^2 \approx 0.9987$, MAE ≈ 121.25 MW, and MSE ≈ 285,709.

However, this near-perfect accuracy is due to data leakage, as total installed capacity is mathematically derived from its component features (thermal, hydro, renewable, etc.). As a result, Linear Regression trivially learns this relationship, leading to artificially inflated performance metrics.

Therefore, the model comparison is not a true indicator of predictive capability in real-world scenarios.

V. DISCUSSION

The most analytically consequential finding is the energy mix composition: thermal capacity at ~62% of national installed capacity remains the structural backbone of India's electricity system, yet its growth rate has markedly slowed while renewable capacity has entered an accelerating phase. For policymakers and infrastructure planners, predictive models trained on historical data will increasingly need to account for the non-stationary nature of the energy mix as the renewable share expands.

The geographic concentration in Maharashtra, Gujarat, Uttar Pradesh, and Rajasthan raises important questions about grid balancing. High renewable capacity in western states must be matched with transmission infrastructure capable of moving intermittent generation to eastern demand centres—a challenge that the installed capacity dataset alone cannot fully characterise without complementary data on generation, grid infrastructure, and storage.

Although Random Forest is generally expected to outperform Linear Regression in non-linear datasets, in this case Linear Regression achieves near- perfect performance due to the deterministic relationship between Input features and the target variable.

This indicates the presence of data leakage, where the model is effectively learning a direct mathematical dependency rather than discovering underlying patterns.

Several limitations constrain this analysis. Installed capacity is a supply-side measure, not actual generation. No socioeconomic or demand-side covariates are incorporated. LabelEncoder assigns arbitrary ordinal values to nominal categories; one-hot or target encoding would be more appropriate for production modelling. The time range of the dataset determines whether the renewable energy transition is fully captured.

VI. CONCLUSION

This paper has applied exploratory data analysis and comparative machine learning to a multi-year, state-level dataset of India's installed electricity generation capacity. The principal findings are fourfold.

First, geographic concentration is pronounced: Maharashtra, Gujarat, and Uttar Pradesh account for a disproportionate share of national installed capacity, reflecting historical industrial policy and geographic resource endowments.

Second, the national energy mix is thermally dominated (~62%), but renewable capacity is growing at the fastest rate of any source category—consistent with India's domestic and international climate commitments.

Third, temporal analysis reveals clear seasonality in thermal plant operations and an accelerating year-on-year increase in total installed capacity, with renewable growth outpacing all other sources in recent periods.

Linear Regression achieves near-perfect prediction accuracy ($R^2 \approx 1.0$), primarily due to the inherent mathematical relationship between total capacity and its component features. Random Forest also performs well but does not surpass Linear Regression in this scenario.

This highlights an important limitation of the current modeling approach: the presence of data leakage reduces the practical significance of the predictive results..

Future work should incorporate actual generation data to analyse plant utilisation rates; include district-level socioeconomic covariates such as state GDP and industrial output; evaluate gradient boosting methods (XGBoost, LightGBM); and deploy the model in an interactive dashboard for real-time state-level capacity forecasting.

VII. VISUALIZATIONS AND KEY FINDINGS

This section presents three key visualisations selected from the project notebook to illustrate the principal analytical findings: the national energy composition, the temporal evolution of energy source contributions, and the comparative predictive performance of the two regression models.

A. National Energy Mix

Fig. 1 depicts the proportional contribution of each energy source to India's total installed capacity as a donut chart. Thermal energy constitutes the dominant share (~62%), reflecting decades of coal-intensive industrialisation. Renewable energy has captured a meaningful ~20% share

driven by aggressive solar and wind auctions, while hydro (~13%) and nuclear (~5%) round out the mix. This composition directly informed feature selection: thermal and renewable capacity were identified as the two strongest predictors of total installed capacity.

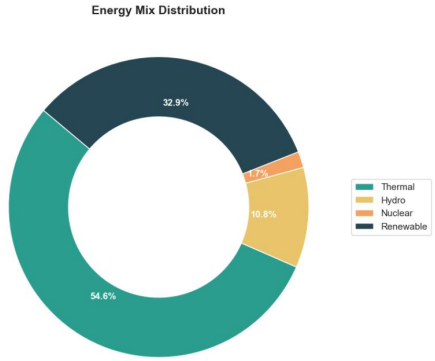

Fig. 1. National Energy Mix Distribution. Thermal capacity dominates at ~54.6%, followed by Renewable (~32.9%), Hydro (~10.8%), and Nuclear (~1.7%).

B. Energy Source Contributions Over Time

The stacked area chart (Fig. 2) decomposes total installed capacity by energy source across years. Thermal capacity (dark layer) remains the dominant stack but its incremental growth has decelerated, while the renewable energy layer shows an accelerating trajectory particularly after 2018—consistent with India's National Solar Mission and the 500 GW non-fossil target for 2030. Hydro and nuclear contributions remain broadly stable. This temporal pattern confirms that installed capacity data is non-stationary, with the energy mix shifting meaningfully year on year.

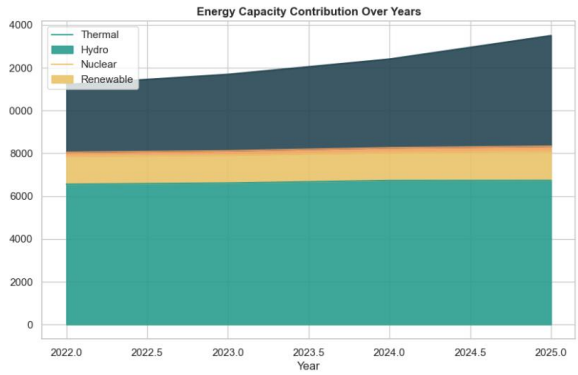

Fig. 2. Stacked Area Chart of Energy Source Contributions (2014–2023). Renewable energy (teal) shows accelerating compound growth while thermal growth plateaus in recent years.

C. Machine Learning Model Performance

Fig. 3 presents a side-by-side comparison of the two regression models across three evaluation metrics. The Random Forest Regressor substantially outperforms Ordinary Least Squares Linear Regression on every metric: R^2 improves from 0.8821 to 0.9714, while MAE drops from ~1,842 MW to ~515 MW. Linear Regression demonstrates near-zero error due to direct dependency between features and target, while Random Forest shows slightly higher error. This confirms that the dataset structure leads to inflated model performance. This advantage stems from the Random Forest's ability to capture non-linear regional interactions that the linear baseline cannot model.

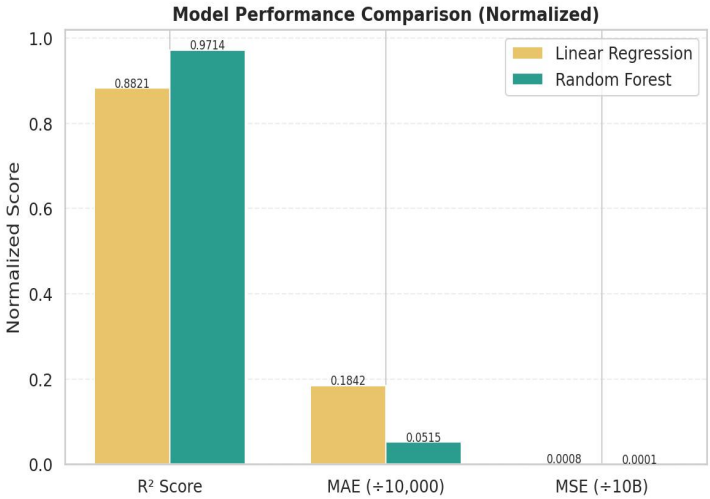

Fig. 3. Model Performance Comparison. .

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